

Using k -Nearest Neighbors Regression to Determine Optimal Shooting Locations in Professional Basketball

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DS 3001

Abstract

Efficiency has become a crucial consideration in today's data-driven sports. In professional (NBA) basketball, it is important to identify locations that maximize the expected number of points for every shot attempt (the *expected value of points earned* or EVPE), taking into account the probability of the shot being successful. We hypothesize that the corners of the three-point line will maximize EVPE, followed by shots near the net and other three-point shots. We use a dataset of NBA shot attempts over a 22-year period and train a k -nearest neighbors regression model to determine which areas on the court maximize EVPE, testing variations of the model that do and do not account for changing play styles over time. We find that the areas immediately around the basket and along the 3-point line of the court yield the highest return in EVPE, validating our hypothesis. There are no significant differences observed between the variations of our model. This information may be useful in determining game strategy for basketball teams, who can target these zones to maximize their scores. Future research may include an exploration of optimal shot locations given specific defensive strategies and usage of image data to analyze defensive coverage on different shots.

Introduction

Data analytics has become a crucial component in professional sports, where players and teams search for every way possible to gain a competitive advantage. The NFL, for instance, hosts an annual Big Data Bowl, whose website touts how it “calls on professional and aspiring amateur

data scientists to devise innovative approaches” to football challenges, complete with a \$100,000 cash prize and presentation opportunities at its annual Scouting Combine. Not to be outdone, the NBA hires data analytics employees; its website states that “our on-and-off-court decisions are driven by Data”.

Data analytics at the NBA team level exploded with the Houston Rockets, whose general manager from 2007 to 2020, Daryl Morey, favored an analytics-heavy approach that prioritized three-point shots and layups over mid-range shots, a philosophy called “Moreyball”. He deemed with the backing of analytics, the mid range shot was not efficient, which revolutionized the league to follow this and start incorporating analytics into their operations. An MIT study published in March 2025 showed that teams who invested in data analytics experienced a positive, statistically significant improvement in wins over the course of twelve years.

For this analysis, we decided to imagine we were working in data analytics for the Washington Wizards, with the goal of finding the best locations for shots. Is Moreyball an effective philosophy, or are there other locations on the court that maximized the points earned by a given shot?

We decided to approach this by defining a new variable: the *expected value of points earned* (EVPE). The EVPE for a certain location is defined as the number of points earned for a given shot from that location, multiplied by the probability of a shot from that location being successful. We aim to find locations on the NBA court that maximize EVPE; our reasoning is that a team that consistently shoots from locations with maximal EVPE will be able to score the most points over the course of a game, putting them in the strongest position to win.

Data

Our data consisted of several Excel sheets retrieved from Kaggle. Each sheet represented one day since 2005 that had NBA games. There were also multiple columns that had to be cleaned up because they had no data. All of the sheets were combined based on their season (ex: one sheet for the 2005-2006 season, one for 2006-2007 season, etc.), in order to avoid excessively opening and closing files, which causes a performance hit. Per-season data files were also chosen because maintaining a single file for the entire dataset would cause that file to exceed GitHub's file size limit.

To group data by season, each row had to look at the date, determine which season it was a part of, and put it in a new “year” column. This was done through making a method called `update_year` and applying it to the column. This process is shown below in Figure 1:

```
def update_year(row):  
    if row['year'] == 2020 and int(row['match_id'][4:6]) >= 12:  
        return row['year'] + 1  
    elif row['year'] != 2020 and 9 <= int(row['match_id'][4:6])  
<= 12:  
        return row['year'] + 1  
    else:  
        return row['year']  
  
df['year'] = df.apply(update_year, axis=1)
```

Figure 1. The `update_year` method.

The rest of the variables were cleared to make sure all the numerical variables were actually numerical. Finally, we saved them into different sheets by season year as mentioned above. The final result is shown in Figure 2:

match_id	shotX	shot Y	quarter	time_rem aining	player	team	made	shot_type	distance	score	opp	status	year
200010310ATL	8.3	20.2	1st quarter	11:29.0	Baron Davis	CHH	FALSE	2-pointer	22	0-2	CHH'	trails	2001
200010310ATL	35.9	15.8	1st quarter	10:41.0	Jamal Mashburn	CHH	FALSE	2-pointer	16	0-2	CHH'	trails	2001
200010310ATL	24	5	1st quarter	10:38.0	Baron Davis	CHH	FALSE	2-pointer	0	0-2	CHH'	trails	2001
200010310ATL	15.4	5.9	1st quarter	09:36.0	Elden Campbell	CHH	FALSE	2-pointer	9	0-4	CHH'	trails	2001
200010310ATL	20.3	25	1st quarter	09:18.0	David Wesley	CHH	TRUE	2-pointer	20	4-2	CHH'	trails	2001

Figure 2. An example of the data used in this study.

The data that is being looked at is NBA shot data from the 2001 season to 2022. It contains information about where the shot is located (both the x and y coordinate), time this is taking place (quarter, time in quarter), the score of the game, and who shot it. This data will be useful for studying how shot behavior has changed over time because we will have every shot attempt from each season over the past 20 years. This will help us track behavior of each season and compare. We also have other variables like time left, and score which help us see how they change shot behavior and how that has changed over time.

Our first question we will try to cover is: how has the location of shot attempts changed over the years? Let's make a scatterplot of shots colored by season.

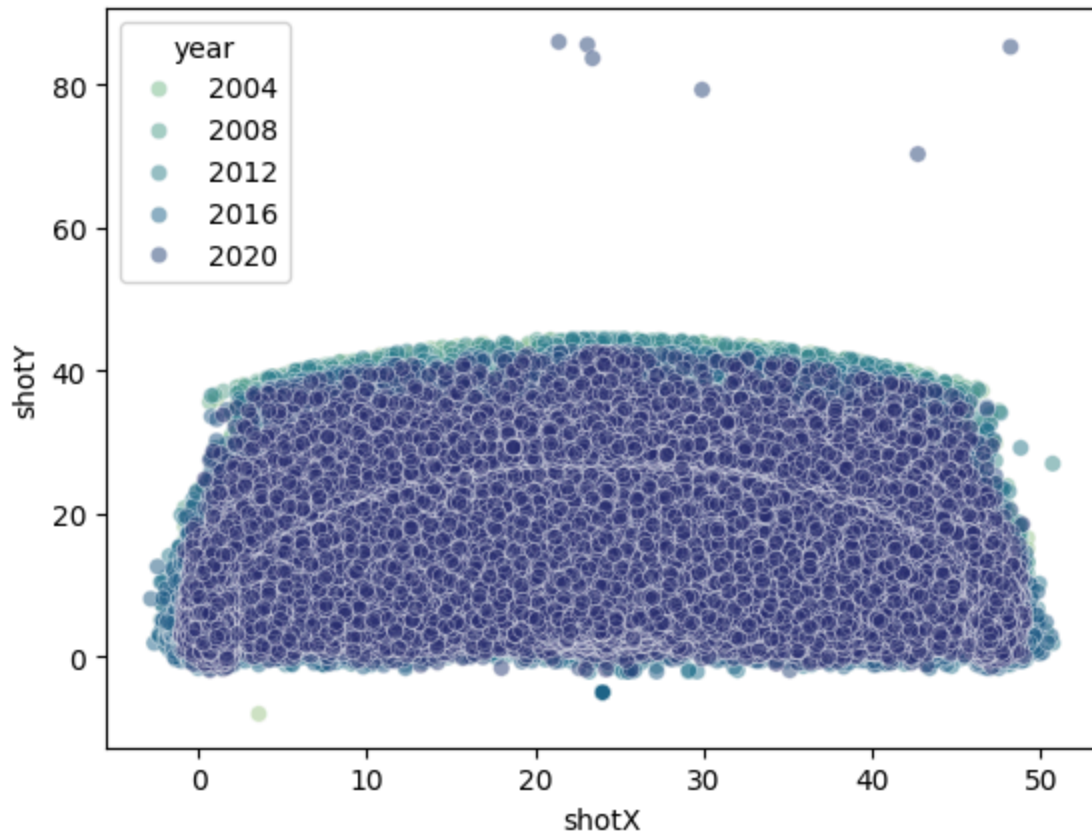


Figure 3. Shots colored by season.

What about whether a shot was successful or not? Are closer shots more successful?

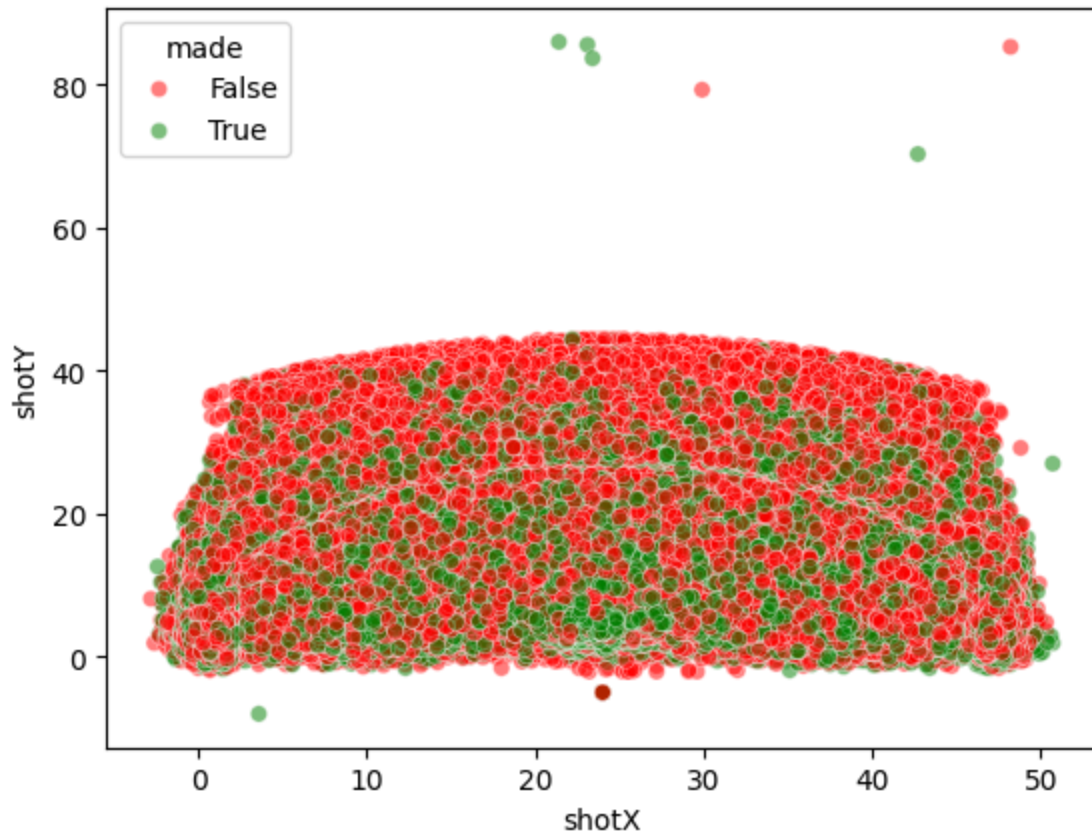


Figure 4. Shots colored by whether they were successful or not.

Is there a tendency for players to shoot closer or farther from the net as the game nears the end?

Let's vary the colors by the total time remaining in the game.

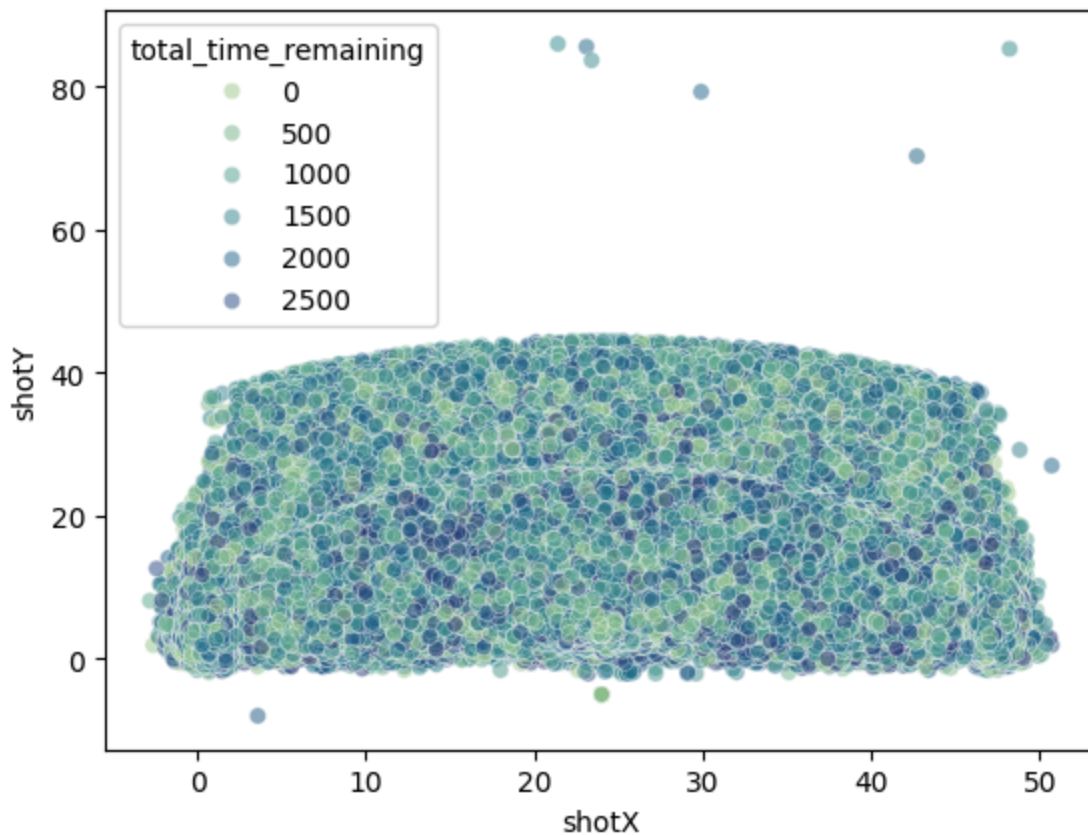


Figure 5. Shots colored by the amount of time remaining in the game when attempted.

The graph shows that farther shots are taken when there is less time remaining. This can be seen as the graph gets lighter the further you go. This would be because in late game situations, threes might be needed because a team is down three points. So, they have to shoot the three or lose.

We can do more with this dataset than just plotting spatial visualizations, though. How does shooting frequency vary with time remaining in the game?

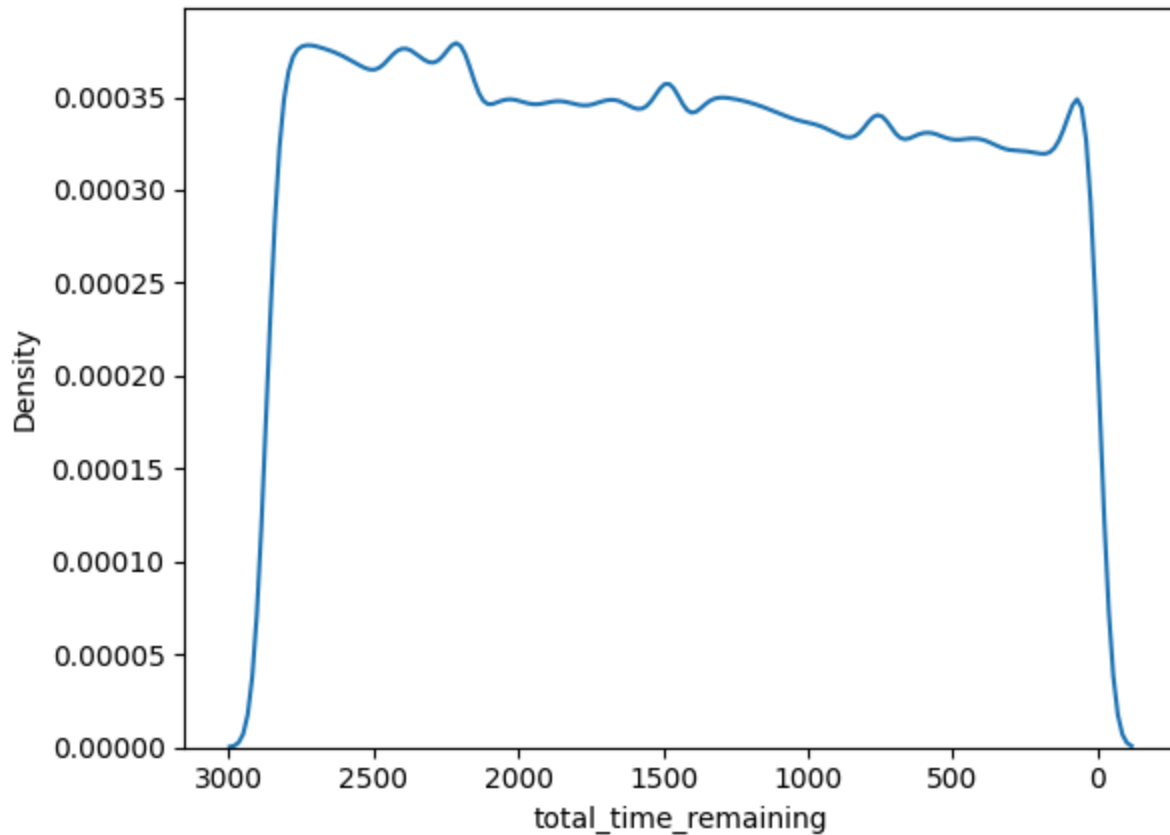


Figure 6. Shooting frequency over the course of a game.

The graph here shows there is a slight decrease in shoot frequency as the game goes on. This could be due to reserves coming in, adjustments in coaching, and fatigue of the players. The spike at the end could be because when teams are down, they have to score quicker.

What about shot attempts over the years? How has that number changed?

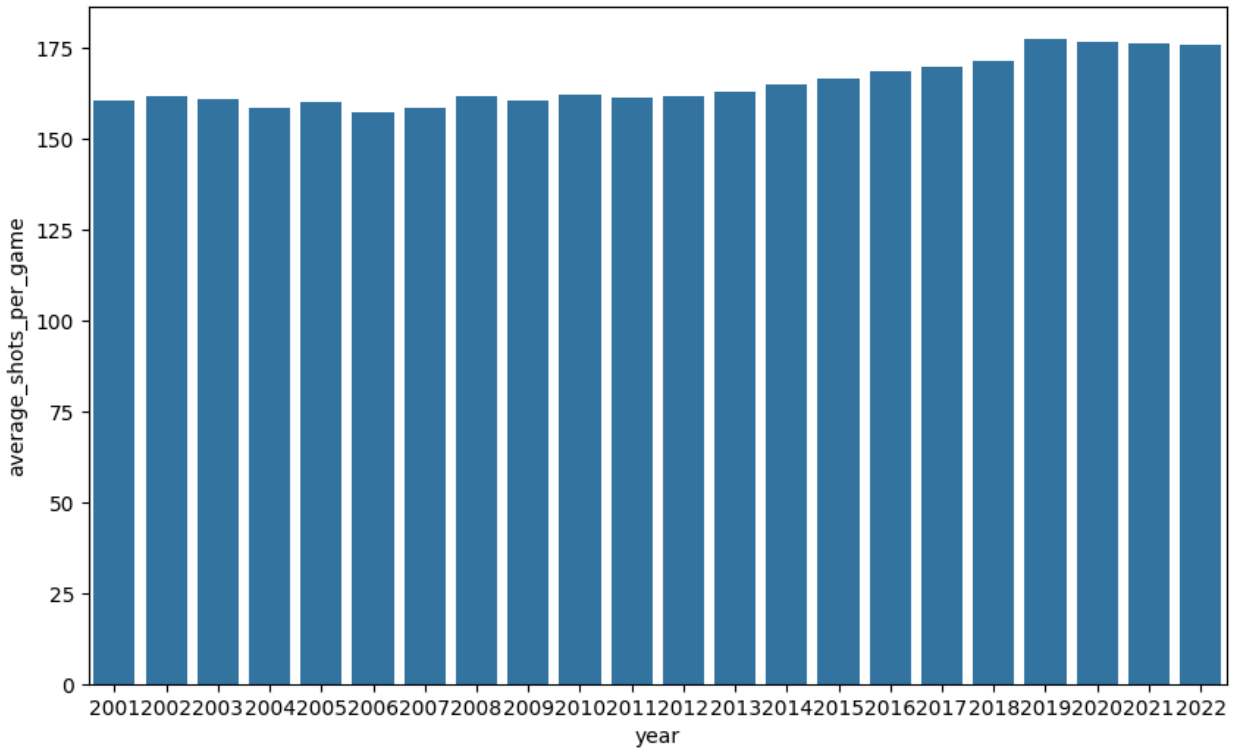


Figure 7. Number of shot attempts over the years.

This graph shows there has been an increase in the shots per year since the early 2000's. This makes sense as the 2000's had a slower pace of play, which was upended by the "Seven Seconds or Less" Phoenix Suns team, which inspired other teams to play faster. In more recent times, we see a higher amount of shots taken.

Do players shoot more 3-pointers when their team is down by a lot, or do they play it safe and try for lots of 2-pointers? For the sake of avoiding outliers, let's look at games where the score difference is no more than 20 points.

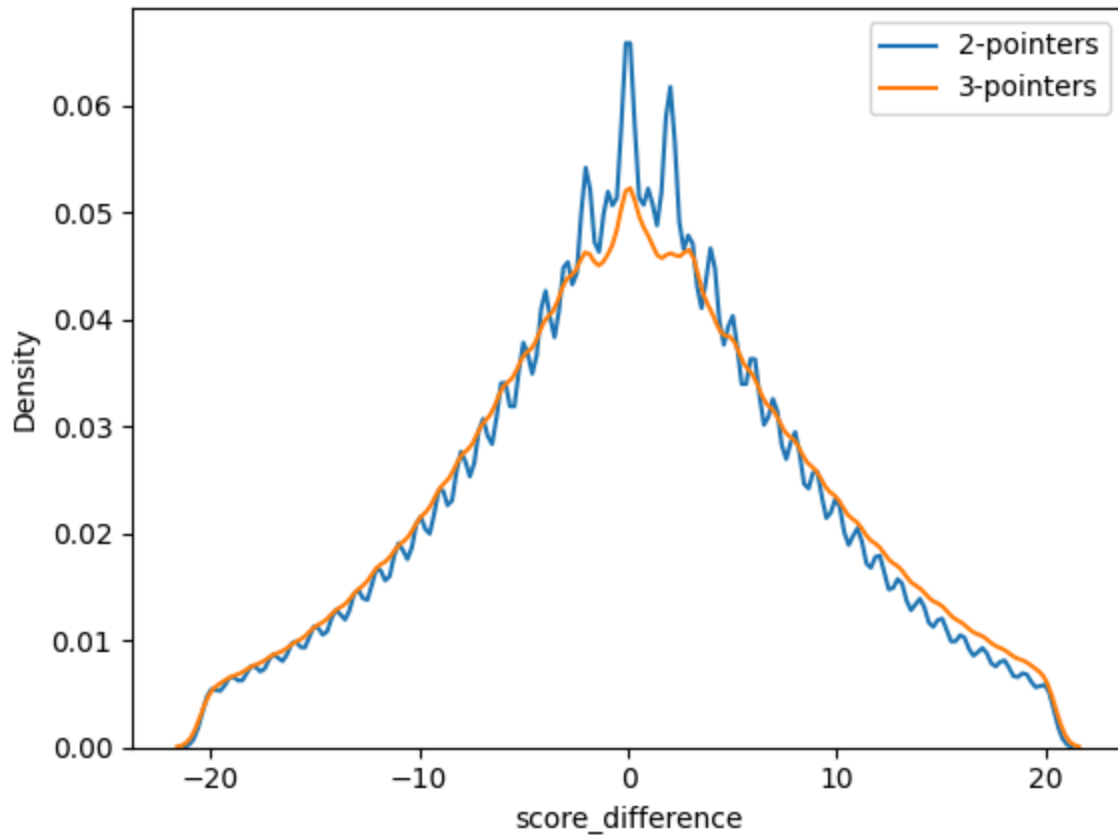


Figure 8. Probability density distribution of shot attempts over a score differential range.

The chart here shows that when the game is close, there are more two point shot attempts than three point shots. This makes sense as if the game is close, then they don't need to risk it for the extra points. However, when down or up by around 6 points, the results get negligible.

Methods

For our analysis, we performed k -nearest neighbors regression (KNN regression), a form of supervised learning, in order to predict the expected point value of a shot at any given location. We split our dataset into 80% training data and 20% testing data. Our model was trained on this data, with the elbow method used to find the optimal value of k – the number of nearest neighbors whose points awarded are averaged in order to find the EVPE.

We expected that a successful model would be able to assign a value between 0 and 3 points (inclusive) for every location on an NBA court, representing that location's EVPE. The distribution would likely vary over the court but be symmetrical in the x dimension across the basket. The model was not intended to predict any values above 2 points for locations where shots can only earn that much.

We anticipated that the time range of our dataset may pose an issue, as the NBA has changed in twenty years and older data may muddle our findings. As a result, we had two variations of our model. The initial version of our model found the k nearest neighbors of a shot given only its coordinates on the court. The second variation included the season of the shot as an additional dimension when determining the distance. All three variables (x -coordinate, y -coordinate, and season) were normalized using max-min normalization.

Results

This paper had the research question: which locations on a basketball court provide the highest EVPE? The hypothesis is that the corners of the three point line will have the highest expected points. Then 0-3 feet shots and other 3-point shots will closely follow. Mid rangers (5 feet - 3-point line) will have the least points expected other than super long range three point shots.

Figure 9 shows our elbow method graph, relating mean squared error with values of k :

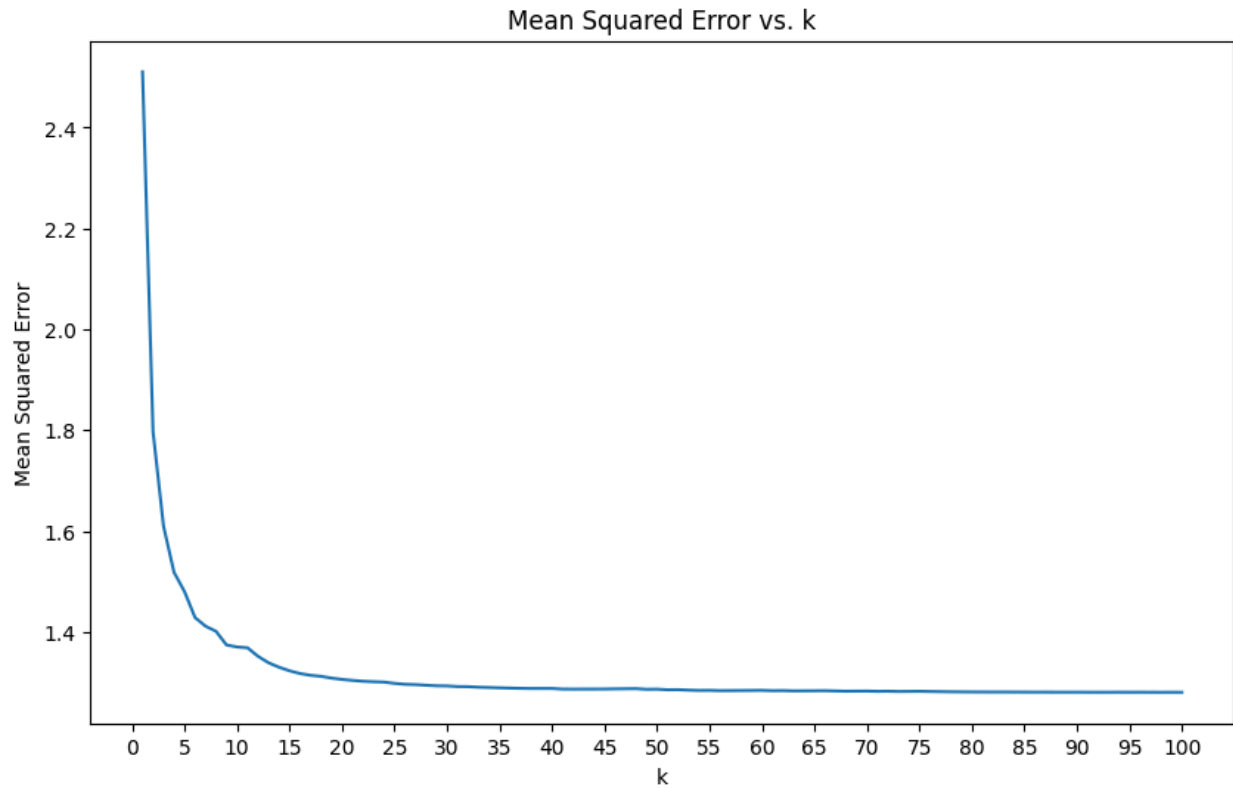


Figure 9. Mean squared error vs. k .

From this, we determined that a k of 8 would be the best choice for our model, as higher values would result in diminishing returns. After this, we ran KNN regression with $k = 8$ to find the EVPE. This used KNN to predict at each spot, if a player took a shot, how many points they would earn. We plotted out the results in a scatter graph:

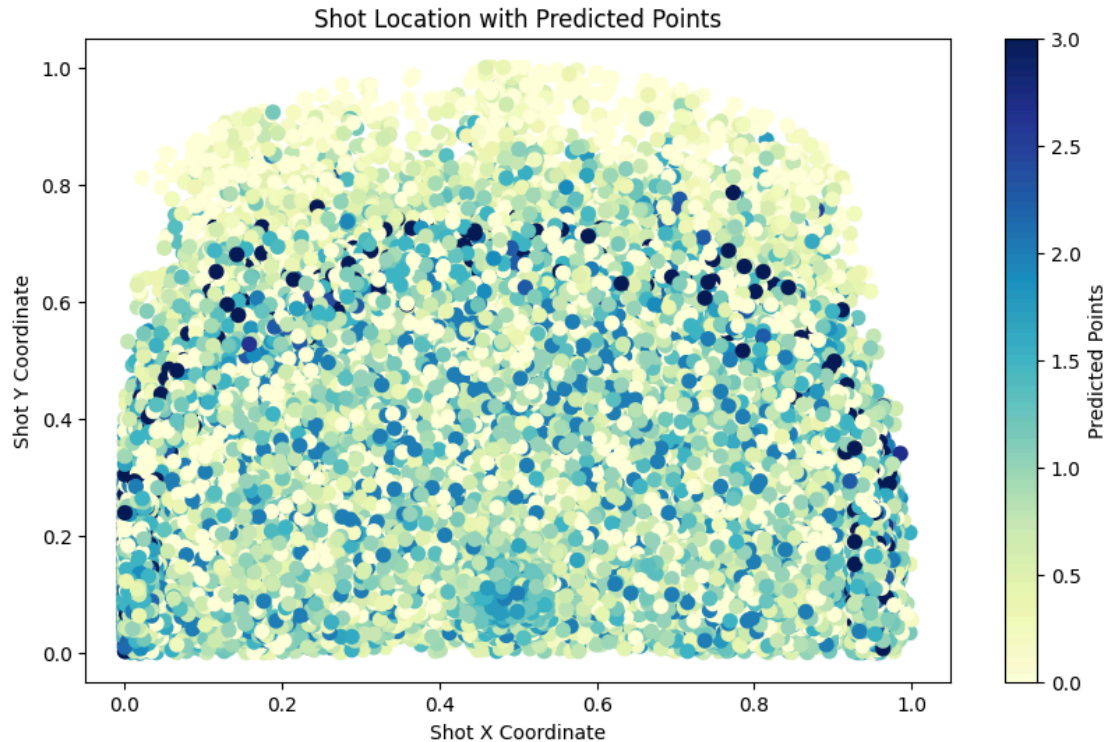


Figure 10. *K*-nearest neighbors regression of EVPE over an NBA court.

This graph shows that the regions with the highest EVPEs are:

- **Right around the basket (around (0.5, 0))** — these shots are layups, where shooting accuracy is of little concern due to how close the player is to the basket.
- **Along the three-point line** (from (0, 0) to about (0, 0.32), then making an arc that peaks around (0.5, 0.64) and curves back to (1, 0.32) before continuing straight down to (1, 0)) — any successful shot on or behind the three-point line can score three points, as opposed to two points for shots inside the three-point line, so shooting along the line allows the player to get as close as possible to the net while still earning the extra point for successful shots.

The mid-range (the area contained by the three-point line and $y = 0$) generally has a lower EVPE associated with it. The lowest EVPE comes from far behind the three point line, where the EVPE is close to zero — these shots are likely deep-3 shots with a low probability of success.

This information is useful, but doesn't account for the fact that play styles have changed over the years in the NBA. Can we adjust our model to include the season, emphasizing the influence of shots taken chronologically close to the one we're predicting? We do that on the graph below:

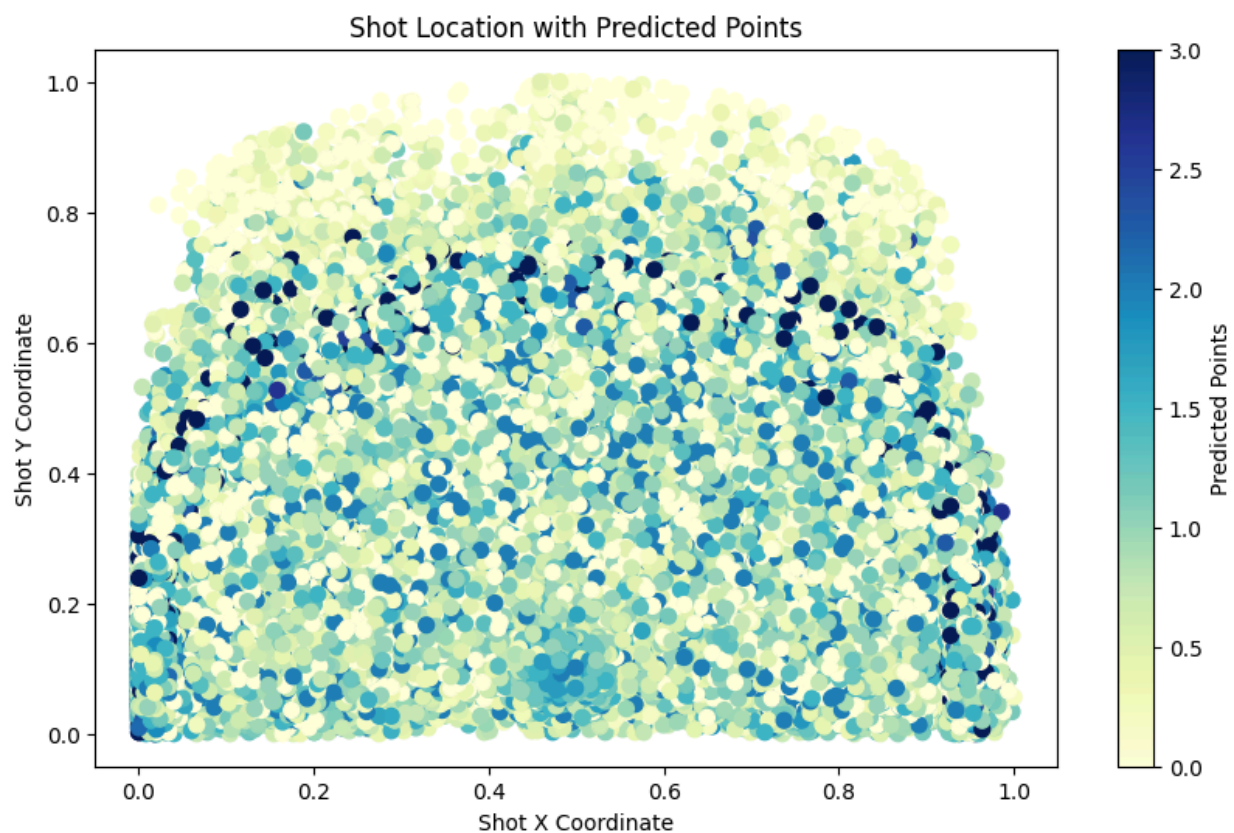


Figure 11. K -nearest neighbors regression of EVPE on an NBA court, adjusted for season proximity.

We find that including the season as a variable (to add more influence to shots taken around the same time as the one being predicted) does not have a significant impact on our

findings. The highest EVPE can still be found immediately around the basket and along the three-point line, while mid-range shots and deep-3 shots are not as effective. We speculate that changing offensive play styles may be balanced out by corresponding changes in defensive play, resulting in a distribution that is relatively stable across the years.

Conclusion

Overall, the model proved that three point shots and layups have the highest EVPE. This would make them the most “efficient” shots as they lead to more points according to our data. So, we’d recommend the Washington Wizards: players shoot three point shots and close shots as they lead to more points. Practices should focus on these shots, while not really practicing mid range two pointers or deep three point shots. If the Wizards incorporate these shots and their players are actually able to properly shoot them, they could possibly have a great offense.

Within our model and conclusion there are some flaws. A potential explanation for the results of our model might involve the defensive pressure on each shot. Areas in the mid-range are often under tighter defensive coverage than those near the rim or beyond the three-point line, resulting in a higher proportion of blocked shots and lower EVPE. However, An uncovered midranger can be more effective than a well defended close shot. Our model and conclusion does not account for that. This means the mid range shot could still have a good reason to be shot. Also, certain regions of the court may exhibit higher EVPEs for different defensive strategies, such as certain zone defense strategies that leave the middle of the court (around $x = 0.5$ in our graphs) open. An additional area of improvement may be to label shot attempts with the active defensive strategy at the time and build separate models for each strategy in order to determine the optimal locations given a specific defense.

Additionally, it is possible that some areas of the court will be so infrequently used for shooting that their EVPEs are unrealistic; for example, only one player might attempt to shoot from a spot, but all of their shots from that location go in, skewing the data.

An extension of the paper would be to focus on the problem of how much each shot is coverage. To do this we would have to find data that would categorize how covered a player is (Wide Open, Open, Lightly Contested, Heavily Contested, and Smothered) on each shot. Then we could convert this into a numerical variable and use it as an additional dimension in the KNN regression model. A more likely plan would be to filter each KNN regression model by how contested the shot is. This would show which shots are best based on how covered the player is. Even if midrangers are less efficient because they are more covered typically, it could be that they are more efficient while covered than the other type of shots. If we cannot find the data of how covered the player is then we'd have to use image data and train a model to identify how covered a shot is.

One limitation we had was the inability to have access to NBA footage. As mentioned in the extension, in the future we'd like to use image data to analyze how covered a player is. The reason we didn't do this in this paper is our limited access to NBA play by play videos. This would cost money to get, but we are not funded. Another problem is the sheer amount of storage and processing power required for it, which we are limited on. We would have to cut down on the amount of seasons we wanted to look at if we did this. So, while this would work for a future extension, we were limited by what we wanted to cover in this project.

Throughout the paper there were some complications we ran into. The main one was how large our data set was. This made it hard to store the data set in one CSV because Github would not allow a file of that size. We did fix this by splitting it up, but anytime we used the whole data

set, we'd have to reattach all the different seasons' data sheets. Again, this took a lot of time due to its size. Another complication was a lack of basketball knowledge. We are data scientists, not basketball coaches who have a great grasp of the game. Everything we do is based on research and our knowledge of the game. The NBA is played on a higher level than anyone else. So, in the future it would be important to get a professional coach to help advise on the project to make sure our project makes sense in the realm of the NBA.

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